

**AMENDMENT NO. 1 JANUARY 2012
TO
IS 800 : 2007 GENERAL CONSTRUCTION IN STEEL —
CODE OF PRACTICE**

(Third Revision)

[(Page (iii), Section 17)] — Insert the following new item, as appropriate:

‘17.15 Bedding Requirement 116’

(Page 6, line 38) — Delete the symbols ‘ C_{my} , C_{mz} ’ and the corresponding explanation.

(Page 8, line 33) — Insert the following symbols and explanations after this line:

‘ K_y , K_z , K_{LT} — Moment amplification factors (see 4.4.2, 4.4.3.1, 4.4.3.3 and 9.3.2.2)’

(Page 18, Table 2, col 2) — Substitute ‘ d/t ’ for ‘ D/t_f ’ for entry against ‘Stem of a T-section, rolled or cut from a rolled I- or H-section’.

(Page 18, Table 2, col 3, 4 and 5) — Substitute ‘but $\geq 42\varepsilon$ ’ for ‘but $\leq 42\varepsilon$ ’ for entries against ‘Web of an I, H or box section’.

(Page 18, Table 2, Notes, last line) — Substitute ‘overall’ for ‘overll’.

(Page 19, Fig. 2, ROLLED CHANNELS) — Substitute ‘d’ for ‘h’ in the figure.

(Page 24, clause 4.4.2, line 10) — Substitute ‘ K_y , K_z ’ for ‘ C_y , C_z ’.

(Page 24, clause 4.4.3.1, line 9) — Substitute ‘ K_y and K_z ’ for ‘ C_y and C_z ’.

(Page 24, clause 4.4.3.3, line 3) — Substitute ‘(K_y , K_z)’ for ‘(C_{my} , C_{mz})’.

(Page 25, clause 4.5.2, line 19) — Insert ‘less’ between ‘be’ and ‘than’.

(Page 31, clause 5.6.1, line 6) — Substitute ‘using load factors of Table 4.’ for ‘using a load factor of 1.0.’

(Page 31, clause 5.6.1, third sentence) — Insert the following at the end:

‘In Table 6, live load should include all post construction loads including superimposed dead loads.’

(Page 33, clause 6.3.3, line 6) — Substitute ‘ $0.9 f_u \gamma_{m0}/f_y \gamma_{m1}$ ’ for ‘ $f_u \gamma_{m0}/f_y \gamma_{m1}$ ’.

(Page 34, clause 7.1.2, line 1) — Substitute the following for the existing:

‘The factored design compression, P in members shall satisfy the following requirement:

$$P < P_d$$

(Page 34, clause 7.1.2.1, line 20) — Substitute ‘ γ_{m0} ’ for ‘ λ_{m0} ’.

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(Page 35, Fig. 8) — Insert ‘ λ ’ as the title of the abscissa (x-axis).

(Page 44, Table 10, col 2, line 3) — Substitute ‘40 mm < $t_f \leq 100$ mm’ for ‘40 ≤ mm < $t_f \leq 100$ mm’.

(Page 45, Table 11, second row, col 1 and 2) — Substitute the following for the existing entries:

(1)	(2)
Restrained	Free

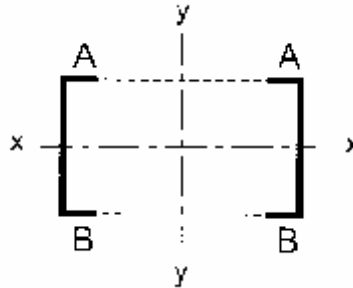
(Page 48, clause 7.5.1.2, line 4) — Add the following in the end:

“, in place of λ in 7.1.2.1 and using curve ‘c’ ($\alpha = 0.49$)”

(Page 48, clause 7.5.1.2, line 9, formula) — Substitute the following for the existing:

$$\lambda_{vv} = \frac{\left(\frac{l}{r_{vv}} \right)}{\varepsilon \sqrt{\frac{\pi^2 E}{250}}} \text{ and } \lambda_{\phi} = \frac{(b_1 + b_2)/2t}{\varepsilon \sqrt{\frac{\pi^2 E}{250}}},$$

(Page 49, Fig. 10) — Substitute the following figure for the existing as appropriate, and substitute ‘Members’ for ‘Numbers’ in the sub-title of Fig. 10C and substitute the existing title of Fig. 10 with ‘TOP RESTRAINT CONDITIONS’:



(Page 53, clause 8.2.1.1, line 3) — Substitute ‘ $d/t_w > 67\varepsilon$ ’ for ‘ $d/t_w \leq 67\varepsilon$ ’.

(Page 54, clause 8.3.1, second para) — Substitute the following for the existing:

‘In simply supported beams with intermediate lateral restraints against lateral torsional buckling, the effective length for lateral torsional buckling, L_{LT} to be used in 8.2.2.1 shall be taken as the length of the relevant segment in between the lateral restraints. In the case of intermediate partial lateral restraints, the effective length, L_{LT} shall be taken as equal to 1.2 times the length of the relevant segment in between the partial lateral restraints.’

(Page 57, Table 14) — Substitute ‘ L_{LT}/r_y ’ and ‘ h/t_f ’ for ‘ KL/r ’ and ‘ h/t_f ’.

(Page 58, clause 8.3.2, line 9) — Insert ‘centre’ between ‘shear’ and ‘and’.

(Page 58, Table 15, col 3, first row) — Substitute ‘Both flanges partially restrained’ for ‘Both flanges fully restrained’.

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(Page 59, clause **8.4.2.1**) — Substitute ' ϵ_w ' for ' ϵ ' and ' f_{yw} ' for ' f_y ', wherever appearing.

(Page 60, clause **8.4.2.2**, col 1, line 18 from top) — Substitute ' $\epsilon_{cr,e}$ ' for ' ϵ_{cr} '.

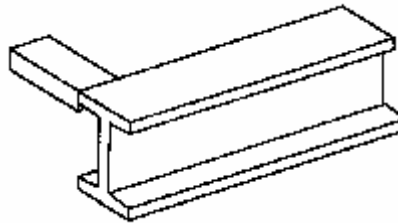
(Page 60, clause **8.4.2.2**, col 2, line 52) — Substitute ' $\text{nearly} = \tan^{-1} \left(\frac{d}{c} \right)$ ' for ' $\text{for} = \tan^{-1} \left(\frac{d}{c} \right)$ '.

(Page 60, clause **8.4.2.2**, col 2, line 55) — Substitute ' $= d \cos \phi - (c - s_c - s_t) \sin \phi$ ' for the existing.

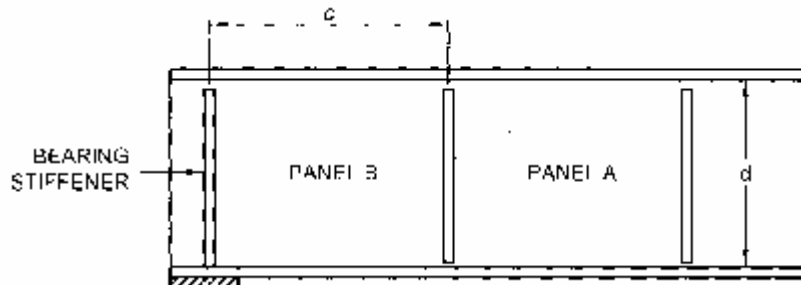
(Page 60, clause **8.4.2.2**, col 2, lines 59 and 60) — Delete the lines.

(Page 60, clause **8.5.1**, line 3) — Insert 'out' between 'carried' and 'in'.

(Page 61, Table 16, last row, col 1) — Substitute the following for the existing figure:



(Page 62, Fig. 12) — Substitute the following for the existing figure:



NOTES

- 1 Panel A is designed utilizing tension field action as given in **8.4.2.2(b)**.
- 2 Panel B is designed using simple post critical method as given in **8.4.2.2(a)**.
- 3 Bearing stiffener is designed for the compressive force due to bearing plus compressive force due to the moment M_{tf} as given in **8.5.3**.

FIG. 12 END PANEL DESIGNED NOT USING TENSION FIELD ACTION

(Page 63, Fig. 13, Notes) — Delete NOTE 2 and renumber the subsequent Note accordingly.

(Page 63, clause **8.6.1.1**) — Substitute ' ϵ_w ' for ' ϵ ' wherever appearing.

(Page 63, clause **8.6.1.1**, line 13) — Substitute ' $c < 0.74d$ ' for ' $c < d$ '.

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(Page 64, clause 8.6.1.2, line 15) — Substitute ‘ ϵ_f = yield stress ratio of flange = $\sqrt{\frac{250}{f_{yf}}}$ ’, for ‘ ϵ_f = yield stress ratio of web = $\sqrt{\frac{250}{f_{yf}}}$ ’.

(Page 65, clause 8.7.1.2, second para, line 1) — Insert ‘stiffener’ between the words ‘web’ and ‘is’.

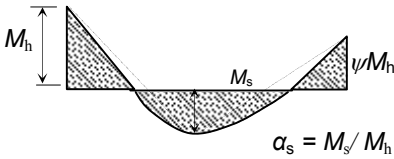
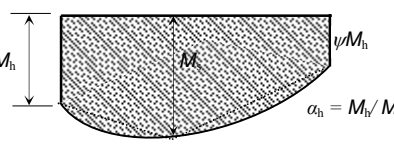
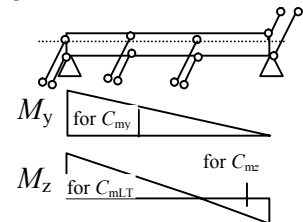
[Page 70, clause 9.3.1.2(c)] — Substitute the following for the existing:

‘c) For standard I or H sections

$$M_{ndz} = 1.11 M_{dz} (1 - n) \leq M_{dz}$$

$$\begin{aligned} \text{for } n \leq 0.2, \quad M_{ndy} &= M_{dy} \\ \text{for } n > 0.2, \quad M_{ndy} &= 1.56 M_{dy} (1 - n) (n + 0.6) \end{aligned}$$

(Page 72, Table 18) — Substitute the following for the existing table:

Bending Moment Diagram (1)	Range (2)		C_{my}, C_{mz}, C_{mLT}	
			Uniform Loading (3)	Concentrated Load (4)
	$-1 \leq \psi \leq 1$		$0.6 + 0.4 \psi \geq 0.4$	
	$0 \leq \alpha_s \leq 1$	$-1 \leq \psi \leq 1$	$0.2 + 0.8 \alpha_s \geq 0.4$	$0.2 + 0.8 \alpha_s \geq 0.4$
	$-1 \leq \alpha_s \leq 0$	$0 \leq \psi \leq 1$	$0.1 - 0.8 \alpha_s \geq 0.4$	$-0.8 \alpha_s \geq 0.4$
$-1 \leq \psi \leq 0$		$0.1(1-\psi) - 0.8 \alpha_s \geq 0.4$	$0.2(-\psi) - 0.8 \alpha_s \geq 0.4$	
		$0 \leq \alpha_h \leq 1$	$-1 \leq \psi \leq 1$	$0.95 - 0.05 \alpha_h$
$-1 \leq \alpha_h \leq 0$		$0 \leq \psi \leq 1$	$0.95 + 0.05 \alpha_h$	$0.90 + 0.10 \alpha_h$
		$-1 \leq \psi \leq 0$	$0.95 + 0.05 \alpha_h (1+2 \psi)$	$0.90 + 0.1 \alpha_h (1+2 \psi)$
For members with sway buckling mode, the equivalent uniform moment factor $C_{my} = C_{mz} = 0.9$.				
C_{my}, C_{mz}, C_{mLT} shall be obtained according to the bending moment diagram between the relevant braced points				
Moment factor	Bending axis	Points braced in direction		
C_{my}	$z-z$	$y-y$		
C_{mz}	$y-y$	$z-z$		
C_{mLT}	$z-z$	$z-z$		

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(Page 75, clause **10.3.2**, line 3) — Substitute the following for the existing:

$$V_{sb} \leq V_{db}$$

(Page 76, clause **10.4.3**, first sentence) — Substitute the following for the existing:

‘Design for friction type bolting, where slip resistance is required at factored design force V_{sf} , shall satisfy the following:’

(Page 76, clause **10.4.3**, line 14) — Substitute ‘ $\mu_f \leq 0.55$ ’ for ‘ $\mu_f = 0.55$ ’.

(Page 76, clause **10.4.3**, Note, line 1) — Substitute ‘ V_{nsf} ’ for ‘ V_{ns} ’.

(Page 77, clause **10.4.5**, col 1, line 6, from top, formula) — Substitute ‘ $0.9 f_{ub} A_n \leq f_{yb} A_{sb} (\gamma_{m1}/\gamma_{m0})$ ’ for ‘ $0.9 f_{ub} A_n \leq f_{yb} A_{sb} (\gamma_{m1}/\gamma_m)$ ’.

(Page 80, clause **10.5.10.2.2**, line 7, formula) — Substitute ‘ f_{br}^2 ’ for ‘ f_{bf}^2 ’.

(Page 89, clause **12.8.2.1**, first sentence) — Substitute the following for the existing:

‘Bracing members shall be made of E250B steel of IS 2062 or of steel having Charpy V-notch energy, $E > 27J$.’

(Page 90, clause **12.11.1**, line 2) — Insert ‘or of steel having Charpy V-notch energy, $E > 27J$ ’ between ‘IS 2062’ and ‘and’.

(Page 106, clause **16.4.1**, line 4, formula) — Substitute ‘ $\frac{f_y(T)}{f_y(20)} = \frac{905 - T}{690} \leq 1.0$ ’ for the existing.

(Page 121, Annex B, clause **B-3.2**, line 10 from top, formula) — Substitute ‘ $\phi_{si} = \frac{\delta_{ui} - \delta_{Li}}{h_i}$ ’ for ‘ $\phi_s = \frac{\delta_u - \delta_L}{h}$ ’.

(Page 121, Annex B, clause **B-3.2**, lines 12, 13 and 16 from top) — Substitute ‘ h_i ’, ‘ δ_{ui} ’ and ‘ δ_{Li} ’ for ‘ h ’, ‘ δ_u ’ and ‘ δ_L ’.

(Page 128, Annex E, clause **E-1.2**, line 5, formula) — Substitute ‘ $(L_{LT})^2$ ’ for ‘ (L_{LT}) ’.

(Page 128, Annex E, clause **E-1.2**, line 30) — Substitute ‘ $(z^2 + y^2)^2$ ’ for ‘ $(z^2 - y^2)$ ’.

(Page 129, Annex E, clause **E-1.2**, col 1, line 18 from top) — Insert ‘St. Venant’s’ before ‘torsion’.

(Page 130, Table 42, col 5, row 7) — Substitute ‘1.267’ for ‘1.257’.

(Page 129, Table 42, col 6, rows 5 and 10) — Substitute ‘1.730’ for ‘1.780’ and ‘1.890’ for ‘1.390’, respectively.

(CED 7)